

Quiz (Habanero)

- Q1.** The temperature in a greenhouse is modeled by the function $f(x) = x^2 + 2x + 1$. If the temperature increases by 3 degrees, the new function is
- (A) $f(x) = x^2 + 2x + 4$
(B) $f(x) = x^2 + 2x + 1 + 3$
(C) $f(x) = x^2 + 2x - 2$
(D) $f(x) = x^2 + 2x$
(E) $f(x) = x^2 + 5x + 4$
- Q2.** The function $g(x) = 2x^2$ represents the growth rate of a particular ecosystem. If the ecosystem's growth rate is reflected over the y-axis, the new function is
- (A) $g(x) = -2x^2$
(B) $g(x) = 2x^2$
(C) $g(x) = 2(-x)^2$
(D) $g(x) = -2(-x)^2$
(E) $g(x) = 2x$
- Q3.** The rainfall in a region is modeled by the function $h(x) = x^2 - 4x + 3$. If the rainfall pattern is shifted 2 units to the right, the new function is
- (A) $h(x) = (x - 2)^2 - 4(x - 2) + 3$
(B) $h(x) = x^2 - 4x + 3$
(C) $h(x) = (x + 2)^2 - 4(x + 2) + 3$
(D) $h(x) = x^2 - 4x - 1$
(E) $h(x) = x^2 - 2x + 1$
- Q4.** The function $f(x) = x^2$ is compressed horizontally by a factor of 2. The new function is
- (A) $f(x) = (2x)^2$
(B) $f(x) = (x/2)^2$
(C) $f(x) = x^2$
(D) $f(x) = 2x^2$
(E) $f(x) = x^2/2$
- Q5.** The temperature in a city is modeled by the function $g(x) = 2x + 1$. If the temperature is reflected over the x-axis, the new function is
- (A) $g(x) = 2x + 1$
(B) $g(x) = -2x - 1$
(C) $g(x) = 2x - 1$
(D) $g(x) = -2x + 1$
(E) $g(x) = x + 2$
- Q6.** The function $h(x) = x^2 - 2x + 1$ is stretched vertically by a factor of 3. The new function is
- (A) $h(x) = 3x^2 - 6x + 3$
(B) $h(x) = x^2 - 2x + 1$
(C) $h(x) = x^2/3 - 2x/3 + 1/3$
(D) $h(x) = 3(x^2 - 2x + 1)$
(E) $h(x) = x^2 - 6x + 3$
- Q7.** The function $f(x) = x^2$ is shifted 1 unit down. The new function is
- (A) $f(x) = x^2 - 1$
(B) $f(x) = x^2 + 1$
(C) $f(x) = (x - 1)^2$
(D) $f(x) = x^2$
(E) $f(x) = x^2 + 2$
- Q8.** The function $g(x) = 2x^2 - 3x + 1$ is compared to its parent function $f(x) = x^2$. The transformation from $f(x)$ to $g(x)$ is
- (A) vertical stretch and horizontal shift
(B) vertical compression and horizontal shift
(C) horizontal stretch and vertical shift
(D) horizontal compression and vertical shift
(E) reflection over x-axis and y-axis

Open Ended Questions (FRQ)

- Q9.** The rainfall in a region is modeled by the function $h(x) = x^2 - 2x + 1$. Describe and apply a vertical shift of 2 units up and a horizontal shift of 1 unit to the right to the function. Show all work and graphs.
- Q10.** The function $f(x) = x^2$ represents the growth rate of a particular ecosystem. Apply a reflection over the y-axis and a vertical compression by a factor of 2 to the function. Compare the transformed graph to the parent function and describe any similarities or differences. Show all work and graphs.

Chef's Tips

- Emphasize the importance of understanding function transformations and their effects on graphs.
- Encourage students to use graphing software or sketch graphs by hand to visualize transformations.
- Remind students to show all work and explain their reasoning for each problem.